Sunzi, Sunzi Suanjing

English Translation

Modern LaTeX Editions of Public-Domain Mathematics Manuscripts

Classical Chinese Mathematics

孫子算經

Sunzi Suanjing

Master Sun’s Mathematical Manual

Bilingual English–Chinese Edition

English Translation • Original Classical Chinese

Attributed to **Sunzi** (Master Sun)

c. 3rd–5th century CE

One of the Ten Computational Canons (算經十書)

Listed during the Tang dynasty (618–907 CE)

Contains the earliest known description of the Chinese Remainder Theorem

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Contents

1 Introduction

關於本書

孫子算經是中國現存最早的數學著作之一，成書於公元三世紀至五世紀之間。唐代將其列為算經十書之一。作者孫子（「孫子」為尊稱）的身份至今仍不確定，但其生活年代遠早於《孫子兵法》的作者孫武。
全書共分三卷（卷）：

- **卷上**：度量衡單位、籌算記數法、乘除算法、九九乘法表
- **卷中**：分數運算、幾何面積與體積計算
- **卷下**：三十六道算題，包括著名的「物不知數」問題——中國剩餘定理的最早記載

About This Book

The *Sunzi Suanjing* (孫子算經, "Master Sun's Mathematical Manual") is one of the earliest surviving Chinese mathematical texts, written during the 3rd to 5th centuries CE. It was listed as one of the *Ten Computational Canons* (算經十書) during the Tang dynasty. The identity of its author, Sunzi ("Master Sun"), remains unknown, though he lived much later than his namesake Sun Tzu, the author of *The Art of War*.
The treatise consists of three volumes (卷):

- **Volume I** (卷上): Units of measurement, counting rod notation, multiplication and division algorithms
- **Volume II** (卷中): Fractions, geometry (areas, volumes), and practical problems
- **Volume III** (卷下): 36 problems including the famous "Problem of the Unknown Quantity"—the **Chinese Remainder Theorem**

歷史意義

孫子算經在數學史上佔有特殊的地位。其最著名的貢獻是**卷下第 26 題**——「物不知數」問題：

> 今有物，不知其數。三、三數之，賸二；五、五數之，賸三；七、七數之，賸二。問物幾何？

此問題提供了公式：

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

其中 r_1, r_2, r_3 分別為以 3、5、7 除後的餘數。

Historical Significance

The *Sunzi Suanjing* holds a special place in the history of mathematics. Its most celebrated contribution is **Problem 26 of Volume III**:

> *"There are certain things whose number is unknown. If we count them by threes, we have two left over; by fives, we have three left over; and by sevens, two are left over. How many things are there?"*

This problem provides the formula:

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

where r_1, r_2, r_3 are the remainders upon division by 3, 5, and 7.

Preface 序

孫子序

孫子曰：夫算者，天地之經緯，群生之元首；五常之本末，陰陽之父母；星辰之建號，三光之表裏；五行之準平，四時之終始；萬物之祖宗，六藝之綱紀。
稽群倫之聚散，考二氣之降升；推寒暑之迭運，步遠近之殊同；觀天道精微之兆基，察地理從橫之長短；采神祇之所在，極成敗之符驗；窮道德之理，究性命之情。
立規矩，準方圓，謹法度，約尺丈，立權衡，平重輕，剖毫釐，析黍絫；歷億載而不朽，施八極而無疆。散之不可勝究，斂之不盈掌握。
嚮之者富有餘，背之者貧且窶；心開者幼沖而即悟，意閉者皓首而難精。夫欲學之者必務量能揆己，志在所專。如是則焉有不成者哉。

Preface by Sunzi

Sun Zi said: Calculation is the meridian and latitude of heaven and earth, the origin and head of all living beings; the root and branch of the Five Constants, the father and mother of yin and yang; the establishment of titles for stars and constellations, the exterior and interior of the Three Lights; the standard balance of the Five Elements, the beginning and end of the Four Seasons; the ancestors of all things, the framework and discipline of the Six Arts.
Investigate the gathering and dispersing of all categories, examine the descending and ascending movements of the Two Qi; track the alternating cycles of cold and heat, pace the differences and similarities between near and far; observe the subtle omens and foundations of the heavenly Dao, examine the lengths and widths of the horizontal and vertical features of geography; gather signs of where gods and spirits reside, thoroughly test omens of success and failure; exhaust the principles of virtue and morality, investigate the nature and essence of life.
Establish rules and compasses, standardize squares and circles, observe laws and regulations carefully, measure with precision in *chi* and *zhang*, set up weights and balances, equalize the heavy and light, dissect thousandths of an inch, analyze grains accumulated one by one; enduring through countless generations without decay, its influence extends to all directions without limit. When dispersed, it is beyond full comprehension; when gathered, it does not fill the palm of one's hand.
Those who face toward it are rich with abundance, while those who turn away from it are poor and destitute; those whose minds open understand even as children, while those whose intentions remain closed find mastery difficult even with a white beard. Therefore, one who wishes to study must necessarily measure his own ability and assess himself carefully, setting his mind on specialization in a particular field. If so, how could there be any failure?

2 Volume I 卷上

1. Length Measurements 度之所起

問 度之所起，起於忽。欲知其忽，蠶所生，吐絲為忽。十忽為一秒，十秒為一毫，十毫為一氂，十氂為一分，十分為一寸，十寸為一尺，十尺為一丈，十丈為一引；五十尺為一端；四十尺為一疋；六尺為一步。二百四十步為一畝。三百步為一里。

Q *The origin of measurement begins with the "hu" (忽). To understand the hu, it is said that silkworms produce it by spinning silk. Ten hu make one miao (秒), ten miao make one hao (毫), ten hao make one li (氂), ten li make one fen (分), ten fen make one cun (寸), ten cun make one chi (尺), ten chi make one zhang (丈), and ten zhang make one yin (引); fifty chi make one duan (端); forty chi make one pi (疋); six chi make one bu (步). Two hundred and forty bu make one mu (畝). Three hundred bu make one li (里).*

2. Weight and Capacity 稱量之所起

問 稱之所起，起於黍。十黍為一絫，十絫為一銖，二十四銖為一兩，十六兩為一斤，三十斤為一鈞，四鈞為一石。量之所起，起於粟。六粟為一圭，十圭為一抄，十抄為一撮，十撮為一勺，十勺為一合，十合為一升，十升為一斗，十斗為一斛。

Q *The origin of weight begins with the grain of millet (黍). Ten millet grains make one lei (絫), ten lei make one zhu (銖), twenty-four zhu make one liang (兩), sixteen liang make one jin (斤), thirty jin make one jun (鈞), and four jun make one dan (石). The origin of capacity begins with the grain of millet. Six millet grains make one gui (圭), ten gui make one chao (抄), ten chao make one cuo (撮), ten cuo make one shao (勺), ten shao make one ge (合), ten ge make one sheng (升), ten sheng make one dou (斗), and ten dou make one hu (斛).*

3. Large Numbers 大數之法

問 凡大數之法，萬萬曰億，萬萬億曰兆，萬萬兆曰京，萬萬京曰陔，萬萬陔曰秭，萬萬秭曰壤，萬萬壤曰溝，萬萬溝曰澗，萬萬澗曰正，萬萬正曰載。

Q *Generally, the method for large numbers: ten thousand times ten thousand is yi (億, 10^8); ten thousand times yi is zhao (兆, 10^{16}); ten thousand times zhao is jing (京, 10^{24}); and so on up to zai (載, 10^{80}).*

4. Geometric Ratios 方圓之比

問 周三徑一。方五邪七；見邪求方，五之，七而一；見方求邪，七之，五而一。

Q *The circumference of a circle is three times its diameter ($\pi \approx 3$). A square with side length five has a diagonal of seven; to find the side from the diagonal, multiply by five and divide by seven; to find the diagonal from the side, multiply by seven and divide by five.*

5. Weights of Materials 物之輕重

問 黃金方寸重一斤。白金方寸重一十四兩。玉方寸重一十二兩。銅方寸重七兩半。鉛方寸重九兩半。鐵方寸重六兩。石方寸重三兩。

Q *One cubic cun of gold weighs one jin. One cubic cun of silver weighs fourteen liang. One cubic cun of jade weighs twelve liang. One cubic cun of copper weighs seven and a half liang. One cubic cun of lead weighs nine and a half liang. One cubic cun of iron weighs six liang. One cubic cun of stone weighs three liang.*

6. Counting Rod Notation 筭位之制

問 凡筭之法，先識其位，一從十橫，百立千僵，千十相望，萬百相當。

籌算記數法： 個位（1--9）縱式；十位（10--90）橫式；百位（100--900）縱式；千位（1000--9000）橫式。縱橫交替，以定位值。

Q *The method for calculation begins by identifying the positions: one is vertical, ten is horizontal; a hundred stands upright and a thousand lies flat.*

Counting Rod Numeral System: Units (1–9): **vertical**; Tens (10–90): **horizontal**; Hundreds (100–900): **vertical**; Thousands (1000–9000): **horizontal**. The pattern alternates.

7. Method of Multiplication 乘法

問 凡乘之法，重置其位。上下相觀，上位有十步至十，有百步至百，有千步至千。以上命下，所得之數列於中位。言十即過，不滿自如。上位乘訖者先去之。下位乘訖者則俱退之。六不積，五不隻。上下相乘，至盡則已。

Q *In multiplication, the positions are arranged again for calculation. By observing vertically and horizontally, if the upper position has ten, it moves to the tens place; if it has a hundred, it moves to the hundreds place; if it has a thousand, it moves to the thousands place. The upper number is used to multiply the lower one, and the resulting product is placed in the middle position. If it reaches ten, carry over; if less than ten, leave as is. After the upper position has been multiplied, remove it first. Once the lower position is multiplied, both are moved back. Six does not accumulate; five does not count as a single unit. Multiplying vertically and horizontally continues until the end is reached.*

8. Method of Division 除法

問 凡除之法，與乘正異。乘得在中央，除得在上方。假令六為法，百為實。以六除百，當進之二等，令在正百下，以六除一，則法多而實少，不可除，故當退就十位。以法除實，言一六而折百為四十，故可除。若實多法少，自當百之，不當復退。故或步法十者置於十位，百者置於百位。上位有空絕者，法退二位。餘法皆如乘時。實有餘者，以法命之，以法為母。實餘為子。

Q *The method of division differs from multiplication. In multiplication, the result is placed in the center; in division, it appears at the top. Suppose six is the divisor (法) and one hundred is the dividend (實). Dividing one hundred by six, it should be advanced two positions and placed under the hundreds place. Dividing one by six results in a larger divisor than dividend, which is not possible; therefore, we must retreat to the tens position. Using the divisor to divide the dividend: saying "one six" and reducing a hundred by sixty results in forty, so division is possible. If the dividend is large and the divisor small, it naturally belongs to the hundreds place, and there is no need to retreat further. Therefore, if a step of ten is taken, it should be placed in the tens position; if a hundred, in the hundreds position. If there is an empty space at the upper position, the divisor must retreat two positions. The remaining methods follow those of multiplication. If there is a remainder in the dividend, name it using the divisor as the denominator. The remainder becomes the numerator.*

9. Example: 81×81 九九八十一自乘

問 九九八十一，自相乘，得幾何?

答 答曰：六千五百六十一。

術 術曰：重置其位，以上八呼下八，八八六十四，即下六千四百於中位。以上八呼下一，一八如八，即於中位下八十。退下位一等，收上位八十。以上位一呼下八，一八如八，即於中位下八十。以上一呼下一，一一如一，即於中位下一。上下位俱收，中位即得六千五百六十一。

Q *Ninety-nine times eighty-one, when multiplied by itself, results in what?*

A *Answer: Six thousand five hundred and sixty-one.*

M *Method: Arrange the positions again, call out eight from above to multiply with eight below; eight eights are sixty-four. Place six thousand four hundred in the middle position. Call out eight from above to one below, one eight is eight; place eighty in the middle position. Move the lower position back by one rank and collect the upper value of eighty. Call out one from above to eight below, one eight is eight, place eighty in the middle position accordingly. Call out one from the upper position to multiply with one from the lower; one times one is one, so place one in the middle position. After collecting both the upper and lower positions, the middle position yields six thousand five hundred sixty-one.*

Modern: $81 \times 81 = (80+1)^2 = 6400+80+80+1 = 6561$.

10. Example: $6561 \div 9$ 九人分之

問 六千五百六十一，九人分之，問人得幾何?

答 答曰：七百二十九。

術 術曰：先置六千五百六十一於中位，為實。下列九人為法。上位置七百，以上七呼下九，七九六十三，即除中位六千三百。退下位一等，即上位置二十。以上二呼下九，二九十八，即除中位一百八十。又更退下位一等，即上位更置九，即以上九呼下九，九九八十一，即除中位八十一。中位盡，收下位。上位所得即人之所得。

Q *Six thousand five hundred sixty-one is to be divided among nine people; how much does each person receive?*

A *Answer: Seven hundred and twenty-nine.*

M *Method: First place six thousand five hundred sixty-one in the middle position as the dividend (實). Place nine people below as the divisor (法). Place seven hundred in the upper position; call out seven from above to multiply by nine below, seven nines are sixty-three, and subtract six thousand three hundred from the middle. Move the lower position back one rank; place twenty in the upper position. Call out two from above to multiply by nine from below, two nines are eighteen, and subtract one hundred eighty from the middle. Move the lower position back another rank, place nine in the upper position again. Call out nine from above to multiply by nine below; nine nines are eighty-one and subtract eighty-one from the middle. When the middle position is exhausted, collect the lower positions. The result in the upper position is what each person receives.*

Modern: $6561 \div 9 = 729$.

3 Volume II 卷中

1. Reducing Fractions 約分

問 今有一十八分之一十二。問約之得幾何? **答** 答曰：三分之二。 **術** 術曰：置十八分在下，一十二分在上。副置二位，以少減多，等數得六，為法。約之，即得。	**Q** *Now there is twelve eighteenths. Ask: reducing it, what does it equal?* **A** *Answer: Two thirds.* **M** *Method: Place eighteen parts below and twelve parts above. Set up both positions, subtract the smaller from the larger; the equal number (等數, GCD) obtained is six, which is the divisor. Reduce by it, and you obtain the answer.* ***Modern:*** $\frac{12}{18} = \frac{12\div6}{18\div6} = \frac{2}{3}$.

2. Grain Conversion 粟米換算

問 今有粟一斗，問為糲米幾何? **答** 答曰：六升。 **術** 術曰：置粟一斗，十升。以糲米率三十乘之，得三百升，為實。以粟率五十為法，除之，即得。	**Q** *Now there is one dou of millet. How much coarse rice does it make?* **A** *Answer: Six sheng.* **M** *Method: Place one dou of millet, which equals ten sheng. Multiply by the coarse rice rate of thirty to get three hundred sheng, as the dividend. Use the millet rate of fifty as the divisor and divide; thus you obtain the answer.* ***Modern:*** $10 \times 30 \div 50 = 6$ *sheng*.

3. Brick Calculation 積磚之數

問 今有屋基南北三丈，東西六丈，欲以磚砌之。凡積二尺，用磚五枚。問計幾何? **答** 答曰：四千五百枚。 **術** 術曰：置東西六丈，以南北三丈乘之，得一千八百尺。以五乘之，得九千尺。以二除之，即得。	**Q** *Now there is a house foundation with a north-south length of three zhang and an east-west width of six zhang, and one wishes to build it using bricks. Generally for every two cubic chi in volume, five bricks are used. Ask how many will be needed?* **A** *Answer: Four thousand five hundred bricks.* **M** *Method: Place the east-west length of six zhang, multiply it by the north-south length of three zhang to get one thousand eight hundred chi. Multiply it by five to get nine thousand chi. Divide it by two, and you will obtain the answer.* ***Modern:*** *Area* $= 30 \times 60 = 1800$ *sq chi. Bricks* $= 1800 \times 5/2 = 4500$.

4. Area of a Circular Field 圓田面積

問 今有圓田周三百步，徑一百步。問得田幾何?

答 答曰：三十一畝奇六十步。

術 術曰：先置周三百步，半之，得一百五十步。又置徑一百步，半之，得五十步。相乘，得七千五百步。以畝法二百四十步除之，即得。

Q *Now there is a round field with circumference three hundred bu, diameter one hundred bu. Ask how much land does it amount to?*

A *Answer: Thirty-one mu and sixty bu extra.*

M *Method: First place the circumference of three hundred bu, halve it to get one hundred fifty bu. Also place the diameter of one hundred bu, halve it, getting fifty bu. Multiply them together to get seven thousand five hundred bu. Divide it by the mu standard of two hundred forty bu, and you will obtain the area.*

Modern: *Area* $\approx \frac{C}{2} \times \frac{d}{2} = 150 \times 50 = 7500$ *sq bu* $= 31$ *mu + 60 bu.*

5. Square Field with Mulberry 方田桑生中央

問 今有方田，桑生中央。從角至桑一百四十七步。問為田幾何?

答 答曰：一頃八十三畝奇一百八十步。

術 術曰：置角至桑一百四十七步，倍之，得二百九十四步。以五乘之，得一千四百七十步。以七除之，得二百一十步。自相乘，得四萬四千一百步。以二百四十步除之，即得。

Q *Now there is a square field with mulberry trees growing in the center. From the corner to the mulberry tree is one hundred forty-seven steps. Ask how much field does it amount to?*

A *Answer: One qing, eighty-three mu and one hundred eighty steps extra.*

M *Method: Place the distance from corner to mulberry tree as one hundred forty-seven bu, double it to get two hundred ninety-four bu. Multiply it by five to get one thousand four hundred seventy steps. Divide it by seven, getting two hundred ten steps. Multiply it by itself to get forty-four thousand one hundred bu. Divide it by two hundred forty steps, and you will get the result.*

Modern: *Diagonal = 147 bu. Using* $\sqrt{2} \approx \frac{7}{5}$, *side* $s = 147 \times 2 \times \frac{5}{7} = 210$ *bu. Area* $= 210^2 = 44100$ *sq bu.*

4 Volume III 卷下

1. Rent Distribution 九家共輸租

問 今有甲、乙、丙、丁、戊、己、庚、辛、壬九家共輸租。甲出三十五斛，乙出四十六斛，丙出五十七斛，丁出六十八斛，戊出七十九斛，己出八十斛，庚出一百斛，辛出二百一十斛，壬出三百二十五斛。凡九家共輸租一千斛。僦運直折二百斛外，問家各幾何?

答 答曰：甲二十八斛。乙三十六斛八蚪。丙四十五斛六蚪。丁五十四斛四蚪。戊六十三斛二蚪。己六十四斛。庚八十斛。辛一百六十八斛。壬二百六十斛。

術 術曰：置甲出三十五斛，以四乘之，得一百四十斛。以五除之，得二十八斛。……

Q *Now there are nine families jointly paying rent: Jia 35 hu, Yi 46 hu, Bing 57 hu, Ding 68 hu, Wu 79 hu, Ji 80 hu, Geng 100 hu, Xin 210 hu, Ren 325 hu. Total 1,000 hu. After deducting transport cost of 200 hu, how much should each family contribute?*

A *Answer: Jia 28 hu, Yi 36 hu 8 dou, Bing 45 hu 6 dou, Ding 54 hu 4 dou, Wu 63 hu 2 dou, Ji 64 hu, Geng 80 hu, Xin 168 hu, Ren 260 hu.*

M *Method: Take each family's contribution, multiply by four, and divide by five.*
Modern: *After 200 hu deduction, 800 hu remains. Each pays* $4/5$ *of original.*

2. Pile of Millet 平地聚粟

問 今有平地聚粟，下周三丈六尺，高四尺五寸。問粟幾何?

答 答曰：一百斛。

術 術曰：置周三丈六尺，自相乘，得一千二百九十六尺。以高四尺五寸乘之，得五千八百三十二尺。以三十六除之，得一百六十二尺。以斛法一尺六寸二分除之，即得。

Q *Now there is a pile of millet on level ground, with base circumference of three zhang six chi and height of four chi five cun. How much millet is there?*

A *Answer: One hundred hu.*

M *Method: Place the circumference of three zhang six chi, multiply it by itself to get 1296 chi. Multiply by the height of 4 chi 5 cun to get 5832 chi. Divide by 36 to get 162 chi. Divide by the hu standard of 1 chi 6 cun 2 fen.*
Modern: *Volume* $\approx \frac{C^2 \times h}{36} = \frac{1296 \times 4.5}{36} = 162$ *cu chi.* $162 \div 1.62 = 100$ *hu.*

3. Go Board 碁局

問 今有碁局方一十九道。問用碁幾何?

答 答曰：三百六十一。

術 術曰：置一十九道，自相乘之，即得。

Q *Now there is a Go board with a square pattern of nineteen lines on each side. How many pieces (intersections) are there?*

A *Answer: Three hundred sixty-one.*

M *Method: Take the nineteen lines, multiply them by themselves, and you get the result.*
Modern: $19^2 = 361$ *intersection points on a Go board.*

4. People and Carts 人與車

問 今有三人共車，二車空；二人共車，九人步。問人與車各幾何?

答 答曰：一十五車。三十九人。

術 術曰：置二車，以三乘之，得六。加步者九人，得車一十五。欲知人者，以二乘車，加九人，即得。

Q *Now there are three people sharing a cart, and two carts remain empty; two people share another cart, while nine people walk. What are the numbers of people and carts?*

A *Answer: Fifteen carts, thirty-nine people.*

M *Method: Place two carts, multiply by three to get six. Add the nine people who walk, getting fifteen carts. To find the number of people, multiply two by the number of carts and add nine people.*

Modern: $3(c-2)=p$ *and* $2c+9=p$, *giving* $c=15$, $p=39$.

5. Guests at a Feast 婦人河上蕩桮

問 今有婦人河上蕩桮。津吏問曰：「桮何以多?」婦人曰：「家有客。」津吏曰：「客幾何?」婦人曰：「二人共飯，三人共羹，四人共肉，凡用桮六十五。不知客幾何?」

答 答曰：六十人。

術 術曰：置六十五桮，以一十二乘之，得七百八十，以十三除之，即得。

Q *Now there is a woman washing cups at the riverbank. The ferry official asked, "Why so many cups?" The woman replied, "There are guests at my home." "How many guests?" "Two people share a meal, three share soup, four share meat, and in total sixty-five cups are used."*

A *Answer: Sixty people.*

M *Method: Take the 65 cups, multiply by twelve to get 780, then divide by thirteen.*

Modern: *Each uses* $\frac{1}{2}+\frac{1}{3}+\frac{1}{4}=\frac{13}{12}$ *cups.* $65 \div \frac{13}{12}=60$.

6. Wood and Rope 木與繩

問　今有木，不知長短。引繩度之，餘繩四尺五寸。屈繩量之，不足一尺。問木長幾何?

答　答曰：六尺五寸。

術　術曰：置餘繩四尺五寸，加不足一尺，共五尺五寸。倍之，得一丈一尺。減餘四尺五寸，即得。

Q *Now there is a piece of wood whose length is unknown. Using a rope to measure it, the rope has 4 chi 5 cun left over. Folding the rope to measure it results in one chi short. What is the length of the wood?*

A *Answer: Six chi and five cun.*

M *Method: Place the excess rope of 4 chi 5 cun, add the deficit of 1 chi, obtaining 5 chi 5 cun. Double it to get 1 zhang 1 chi. Subtract the excess of 4 chi 5 cun.*
Modern: $R = W + 4.5$ *and* $R/2 = W - 1$. *Solving:* $W = 6.5$ *chi.*

7. Rice in a Vessel 器中米

問　今有器中米，不知其數。前人取半，中人三分取一，後人四分取一，餘米一斗五升。問本米幾何?

答　答曰：六斗。

術　術曰：置餘米一斗五升，以六乘之，得九斗。以二除之，得四斗五升。以四乘之，得一斛八斗。以三除之，即得。

Q *Now there is rice in a vessel, but its quantity is unknown. The first person took half, the second took one-third of what remained, the third took one-fourth of what was left, and there were 1 dou 5 sheng remaining. What was the original amount?*

A *Answer: Six dou.*

M *Method: Place the remaining rice of 1 dou 5 sheng, multiply by six to get 9 dou. Divide by two to get 4 dou 5 sheng. Multiply by four to get 1 hu 8 dou. Divide by three.*
Modern: $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times x = \frac{x}{4} = 1.5$ *dou. So* $x = 6$ *dou.*

8. Birds and Beasts 獸六首四足

問　今有獸六首四足，禽四首二足。上有七十六首，下有四十六足。問禽、獸各幾何?

答　答曰：八獸，七禽。

術　術曰：倍足以減首，餘，半之，即禽。四乘禽數，減足，餘，半之，即獸。

Q *Now there are beasts with six heads and four feet, and birds with four heads and two feet. Above there are seventy-six heads, below there are forty-six feet. Ask: How many birds and beasts are there?*

A *Answer: Eight beasts, seven birds.*

M *Method: Double the feet and subtract the heads; halve the remainder to get the number of birds. Multiply the number of birds by four, subtract from the feet; halve the remainder to get the number of beasts.*
Modern: $6b + 4i = 76$, $4b + 2i = 46$. *Solving:* $b = 8$, $i = 7$.

9. One Hundred Deer 百鹿入城

問 今有百鹿入城，家取一鹿，不盡；又三家共一鹿，適盡。問城中家幾何?

答 答曰：七十五家。

術 術曰：以盈不足取之。假令七十二家，鹿不盡四。令之九十家，鹿不足二十。置七十二於右上，盈四於右下。置九十於左上，不足二十於左下。維乘之，所得，并為實。并盈不足為法。除之，即得。

Q *Now there are one hundred deer entering a city; each household takes one deer, but they do not finish; then three households share one deer and it is just finished. Ask: How many households are in the city?*

A *Answer: Seventy-five households.*

M *Method: Use the surplus-and-deficit method. Suppose 72 households: 4 deer remain (surplus). Suppose 90: 20 deer short (deficit). Cross-multiply and add to get the dividend. Add surplus and deficit to get the divisor.*

Modern: *$h + h/3 = 100$, so $4h/3 = 100$, giving $h = 75$.*

10. Pheasants and Rabbits 雉兔同籠

問 今有雉、兔同籠，上有三十五頭，下九十四足。問雉、兔各幾何?

答 答曰：雉二十三。兔一十二。

術 術曰：上置三十五頭，下置九十四足。半其足，得四十七。以少減多，再命之，上三除下三，上五除下五。下有一除上一，下有二除上二，即得。
又術曰：上置頭，下置足。半其足，以頭除足，以足除頭，即得。

Q *Now there are pheasants and rabbits in a cage, with thirty-five heads above and ninety-four feet below. Ask: How many pheasants and how many rabbits are there?*

A *Answer: Twenty-three pheasants, twelve rabbits.*

M *Method: Place thirty-five heads above, ninety-four feet below. Halve the feet to get forty-seven. Subtract the smaller from the larger, and command it again.*
Alternative method: Place heads above, feet below. Halve the feet, use heads to remove feet and feet to remove heads.
Modern: $p + r = 35$, $2p + 4r = 94$. *Solving:* $p = 23$, $r = 12$.

11. Nine Dikes 出門望見九隄

問 今有出門望見九隄。隄有九木，木有九枝，枝有九巢，巢有九禽，禽有九雛，雛有九毛，毛有九色。問各幾何?

答 答曰：木八十一。枝七百二十九。巢六千五百六十一。禽五萬九千四十九。雛五十三萬一千四百四十一。毛四百七十八萬二千九百六十九。色四千三百四萬六千七百二十一。

術 術曰：置九隄，以九乘之，得木之數。又以九乘之，得枝之數。又以九乘之，得巢之數。又以九乘之，得禽之數。又以九乘之，得雛之數。又以九乘之，得毛之數。又以九乘之，得色之數。

Q *Now there is someone who goes out of the gate and sees nine dikes. Each dike has nine trees, each tree has nine branches, each branch has nine nests, each nest has nine birds, each bird has nine chicks, each chick has nine feathers, and each feather has nine colors. Ask: How many of each are there?*

A *Answer: 81 trees, 729 branches, 6,561 nests, 59,049 birds, 531,441 chicks, 4,782,969 feathers, 43,046,721 colors.*

M *Method: Place nine dikes, multiply by nine to get the number of trees. Multiply by nine again for each successive level.*
Modern: *Powers of nine:* $9^2 = 81$, $9^3 = 729$, $9^4 = 6561$, $9^5 = 59049$, $9^6 = 531441$, $9^7 = 4782969$, $9^8 = 43046721$.

12. Three Daughters 三女歸家

問 今有三女，長女五日一歸，中女四日一歸，少女三日一歸。問三女幾何日相會?

答 答曰：六十日。

術 術曰：置長女五日、中女四日、少女三日，於右方。各列一算於左方。維乘之，各得所到數：長女十二到，中女十五到，少女二十到。又各以歸日乘到數，即得。

Q *Now there are three daughters: the eldest returns home every five days, the middle every four days, and the youngest every three days. Ask: After how many days will the three daughters meet again?*

A *Answer: Sixty days.*

M *Method: Place the eldest daughter's five days, the middle daughter's four days, and the youngest daughter's three days on the right side. Place one counter for each on the left side. Multiply them, and you get their respective numbers of arrivals: twelve, fifteen, and twenty. Then multiply each return day by the number of arrivals.*
Modern: *This is the least common multiple:* *lcm*$(3, 4, 5) = 60$.

物不知數 — The Chinese Remainder Theorem

問 今有物，不知其數。三、三數之，賸二；五、五數之，賸三；七、七數之，賸二。問物幾何?

答 答曰：二十三。

術 術曰：「三、三數之，賸二」，置一百四十；「五、五數之，賸三」，置六十三；「七、七數之，賸二」，置三十。并之，得二百三十三。以二百一十減之，即得。
凡三、三數之，賸一，則置七十；五、五數之，賸一，則置二十一；七、七數之，賸一，則置十五。一百六以上，以一百五減之，即得。

Q *Now there is an object, but we do not know its quantity. Counting it in groups of three leaves a remainder of two; counting it in groups of five leaves a remainder of three; counting it in groups of seven leaves a remainder of two. What is the quantity of the object?*

A *Answer: Twenty-three.*

M *Method: "Counting in groups of three, remainder two," set down 140; "Counting in groups of five, remainder three," set down 63; "Counting in groups of seven, remainder two," set down 30. Add them: 233. Subtract 210, and you will obtain the answer.*
Generally, if counting in groups of three leaves remainder one, set down 70; if counting in groups of five leaves remainder one, set down 21; if counting in groups of seven leaves remainder one, set down 15. If the total is more than 160, subtract 150; thus you obtain the answer.

Modern Mathematical Analysis 現代數學分析

以現代符號表示，此問題要求找出最小的正整數 N，滿足以下同餘方程組：

$$\begin{aligned} N &\equiv 2 \pmod 3 \\ N &\equiv 3 \pmod 5 \\ N &\equiv 2 \pmod 7 \end{aligned}$$

解法給出：

$$\begin{aligned} N &= 140 + 63 + 30 - 210 \\ &= 233 - 210 \\ &= 23 \end{aligned}$$

三個關鍵數字 **70**、**21**、**15** 有特殊性質：

- $70 = 2 \times 5 \times 7 \equiv 1 \pmod 3$
- $21 = 1 \times 3 \times 7 \equiv 1 \pmod 5$
- $15 = 1 \times 3 \times 5 \equiv 1 \pmod 7$

由於 $3 \times 5 \times 7 = 105$，一般解為：

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

其中 r_1, r_2, r_3 為餘數，k 為使 N 為最小正整數的整數。

In modern notation, this problem asks us to find the smallest positive integer N satisfying:

$$\begin{aligned} N &\equiv 2 \pmod 3 \\ N &\equiv 3 \pmod 5 \\ N &\equiv 2 \pmod 7 \end{aligned}$$

The method gives:

$$\begin{aligned} N &= 140 + 63 + 30 - 210 \\ &= 233 - 210 \\ &= 23 \end{aligned}$$

The three key numbers **70**, **21**, and **15** have special properties:

- $70 = 2 \times 5 \times 7 \equiv 1 \pmod 3$
- $21 = 1 \times 3 \times 7 \equiv 1 \pmod 5$
- $15 = 1 \times 3 \times 5 \equiv 1 \pmod 7$

Since $3 \times 5 \times 7 = 105$, the general solution is:

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

where r_1, r_2, r_3 are the remainders and k is chosen to make N positive and minimal.

Verification 驗算

$$\begin{aligned} 23 \div 3 &= 7 \text{ 餘}\mathbf{2} \quad \checkmark \\ 23 \div 5 &= 4 \text{ 餘}\mathbf{3} \quad \checkmark \\ 23 \div 7 &= 3 \text{ 餘}\mathbf{2} \quad \checkmark \end{aligned}$$

孫子公式：

$$\boxed{N = 70 \times r_1 + 21 \times r_2 + 15 \times r_3 - 105 \times k}$$

$$\begin{aligned} 23 \div 3 &= 7 \text{ remainder } \mathbf{2} \quad \checkmark \\ 23 \div 5 &= 4 \text{ remainder } \mathbf{3} \quad \checkmark \\ 23 \div 7 &= 3 \text{ remainder } \mathbf{2} \quad \checkmark \end{aligned}$$

Sunzi's Formula:

$$\boxed{N = 70 \times r_1 + 21 \times r_2 + 15 \times r_3 - 105 \times k}$$

Historical Significance 歷史意義

此問題被稱為「物不知數」，是**中國剩餘定理**的最早記載。雖然孫子的文本只給出了具體的數值方法而沒有證明，但它包含了核心思想：

1. 找到一個數，它是除一個模數外所有模數的倍數，且除以該模數餘一
2. 將每個這樣的數乘以相應的餘數
3. 將結果相加，減去最小公倍數的倍數，得到最小正解

一般定理後來由**秦九韶**（1202–1261）在其《數書九章》（1247 年）中完整發展，稱之為「大衍求一術」。

This problem, known as the **"Problem of the Unknown Quantity"** (物不知數), is the earliest known statement of the **Chinese Remainder Theorem**. Sunzi's text gives the essential idea:

1. Find numbers that are multiples of all but one modulus, leaving remainder 1 when divided by the remaining modulus
2. Multiply each such number by the corresponding remainder
3. Sum the results and subtract multiples of the LCM to get the smallest positive solution

The general theorem was later fully developed by **Qin Jiushao** (1202–1261) in his *Mathematical Treatise in Nine Sections* (數書九章, 1247 CE).

The "Sunzi Song" 孫子歌

一首總結此方法的韻文廣泛流傳，後由程大位收錄於《算法統宗》（1592 年）：

三人同行七十稀，
五樹梅花廿一枝，
七子團圓正半月，
除百零五便得知。

關鍵數字：**70**（模 3）、**21**（模 5）、**15**（模 7）、**105**（最小公倍數）。

A rhymed verse summarizing the method circulated widely and was later recorded by Cheng Dawei in his *Systematic Treatise on Algorithms* (算法統宗, 1592):

"Three people walking together—seventy is rare;
Five plum trees—twenty-one branches;
Seven sons reunite—half a full moon (15);
Subtract one hundred and five, then you will know."

Key numbers: **70** for mod 3, **21** for mod 5, **15** for mod 7, **105** as the LCM.

Remaining Problems from Volume III 卷下餘題

13. Money Exchange 二人持錢

問 今有甲、乙二人，持錢各不知數。甲得乙中半，可滿四十八。乙得甲太半，亦滿四十八。問甲、乙二人元持錢各幾何?

答 答曰：甲持錢三十六。乙持錢二十四。

術 術曰：如方程求之。置二甲、一乙、錢九十六，於右方。置二甲、三乙、錢一百四十四，於左方。以右方二乘左方……得二十四錢，為乙錢。……得三十六，為甲錢也。

Q *Now there are two people, Jia and Yi, each holding an unknown amount of money. If Jia receives half of what Yi has, he will have forty-eight. If Yi receives two-thirds of what Jia has, he too will have forty-eight. Ask: Originally, how much did Jia and Yi each hold?*

A *Answer: Jia originally held thirty-six units of money. Yi originally held twenty-four units.*

M *Method: Solve as a system of linear equations. Place 2 Jia, 1 Yi, 96 coins on the right. Place 2 Jia, 3 Yi, 144 coins on the left. Cross-multiply to obtain twenty-four coins as Yi's amount and thirty-six as Jia's amount.*
Modern: $J+Y/2 = 48$, $Y+2J/3 = 48$. *Solving:* $J = 36$, $Y = 24$.

14. Three Chickens 三雞共啄粟

問 今有三雞共啄粟一千一粒。雛啄一，母啄二，翁啄四。主責本粟，三雞主各償幾何?

答 答曰：雞雛主一百四十三。雞母主二百八十六。雞翁主五百七十二。

術 術曰：置粟一千一粒，為實。副并三雞所啄粟七粒，為法。除之，得一百四十三粒，為雞雛主所償之數。遞倍之，即得母、翁主所償之數。

Q *Now there are three chickens pecking at one thousand and one grains of millet. A chick pecks one grain, a hen pecks two, and a rooster pecks four. How many grains should each owner compensate?*

A *Answer: The chick's owner compensates 143, the hen's owner compensates 286, and the rooster's owner compensates 572.*

M *Method: Place 1001 grains of millet as the dividend. Add together the three chickens' consumption of seven grains as the divisor. Divide: 143 grains is the chick owner's compensation. Double successively to get the hen's and rooster's owner's compensation.*
Modern: *Ratio is* $1:2:4$. $1001 \div 7 = 143$. *So:* $143, 286, 572$.

15. Fish in a Ditch 九里渠

問 今有九里渠，三寸魚，頭頭相次。問魚得幾何?

答 答曰：五萬四千。

術 術曰：置九里，以三百步乘之，得二千七百步。又以六尺乘之，得一萬六千二百尺。上十之，得一十六萬二千寸。以魚三寸除之，即得。

Q *Now there is a ditch nine li long; fish of three cun are lined up head to tail. Ask: How many fish can fit?*

A *Answer: Fifty-four thousand.*

M *Method: Place nine li, multiply by three hundred bu to get two thousand seven hundred bu. Multiply by six chi to get sixteen thousand two hundred chi. Convert upward by ten to get one hundred sixty-two thousand cun. Divide by the fish length of three cun.*
Modern: *9 li* $= 9 \times 300 \times 6 = 16200$ *chi* $= 162000$ *cun.* $162000/3 = 54000$ *fish.*

16. Wheel Turns 輪匝

問 今有長安、洛陽相去九百里。車輪一匝一丈八尺。欲自洛陽至長安，問輪匝幾何?

答 答曰：九萬匝。

術 術曰：置九百里，以三百步乘之，得二十七萬步。又以六尺乘之，得一百六十二萬尺。以車輪一丈八尺為法。除之，即得。

Q *Now there are Chang'an and Luoyang, 900 li apart. A cart wheel turns one revolution for every one zhang eight chi. If one wishes to travel from Luoyang to Chang'an, how many wheel turns are needed?*

A *Answer: Ninety thousand turns.*

M *Method: Place nine hundred li, multiply by three hundred bu to get two hundred seventy thousand bu. Multiply by six chi to get one million six hundred twenty thousand chi. Use the wheel circumference of one zhang eight chi as the divisor.*
Modern: *Distance* $= 900 \times 300 \times 6 = 1{,}620{,}000$ *chi. Wheel* $= 18$ *chi. Turns* $= 1{,}620{,}000/18 = 90{,}000$.

17. Predicting Gender 孕婦行年

問 今有孕婦行年二十九，難九月。未知所生?

答 答曰：生男。

術 術曰：置四十九，加難月，減行年。所餘，以天除一，地除二，人除三，四時除四，五行除五，六律除六，七星除七，八風除八，九州除九。其不盡者，奇則為男，耦則為女。

Q *Now there is a pregnant woman whose age is twenty-nine, giving birth in the ninth month. It is unknown what will be born?*

A *Answer: A boy will be born.*

M *Method: Set forty-nine, add the month of childbirth, subtract the age. The remainder is divided by: Heaven 1, Earth 2, Humanity 3, Four Seasons 4, Five Elements 5, Six Laws 6, Seven Stars 7, Eight Winds 8, Nine Provinces 9. If the remainder is odd, a boy; if even, a girl.*

Epilogue 跋

孫子算經是古代中國數學高度發展的非凡見證。雖然本文涵蓋了基本算術——度量衡單位、乘法表和簡單分數——但其最偉大的遺產在於簡短而精妙的「物不知數」問題，它為現代數論開啟了大門。

中國剩餘定理——這一方法在西方後來的稱呼——要過了一千多年才在歐洲被完全形式化。孫子所描述的算法包含了現代定理的基本要素：構造具有特定整除性質的輔助數字，將它們與給定的餘數結合，以及對最小公倍數取模的約簡。

這部著作的影響遠遠超出了中國。這個問題在整個東亞作為數學難題流傳，而秦九韶所發展的一般方法代表了中世紀世界數學最偉大的成就之一。

The *Sunzi Suanjing* stands as a remarkable testament to the sophistication of ancient Chinese mathematics. While the text covers basic arithmetic—units of measurement, multiplication tables, and simple fractions—its greatest legacy lies in the brief but brilliant "Problem of the Unknown Quantity" that opens the door to modern number theory.

The Chinese Remainder Theorem, as this method came to be known in the West, would not be fully formalized in Europe for over a millennium. The algorithm described by Sunzi contains the essential ingredients of the modern theorem: the construction of auxiliary numbers with specific divisibility properties, their combination with the given remainders, and reduction modulo the least common multiple.

The influence of this text extended far beyond China. The problem circulated as a mathematical puzzle throughout East Asia, and the general method developed by Qin Jiushao represents one of the great achievements of medieval mathematics anywhere in the world.

Bilingual edition typeset from the classical Chinese text
as preserved in the Chinese Text Project (`https://ctext.org/sunzi-suan-jing`)